QP Code: 25900748

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS)
REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCBTS100 - FOOD, NUTRITION AND MEDICINE (2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Part A
Multiple Choice Questions
Answer **All** Questions
Each Question carries **1** mark

1.Niacin can be synthesized in human beings from [U]/ [CO2,CO3]

Answer: d) Tryptophan

2.A diabetic patient is advised to avoid simple sugars. Which of the following would be a better alternative for maintaining blood sugar levels? [A] [CO1,CO2]

Answer: b) Whole grains and legumes

3.Which of the following types of functional food can help support immune system?

Answer: b) Fermented foods [K] [CO2]

4. The current rates of disease sare increasing at alarming speeds and mostare closely linked with known [K] / [CO2,CO3,CO4]

Answer: a) Carcinogens

5.Probiotics are often recommended to help with which of the following conditions [U] [CO1]

Answer: d)None of the above

(5 x 1 = 5)

Part B
Short Answer Questions
Answer any **Five** questions,
Each question carries **2** marks

6.Explain the term balanced diet and its significance. [U] [CO1,CO2]

A balanced diet provides the body with essential nutrients like carbohydrates, proteins, fats, vitamins, minerals, fibre, and water in the correct proportions to function properly. Its significance lies in maintaining overall health by supporting growth, aiding daily functions, boosting the immune system, and preventing chronic diseases.

7.State two health risks of excessive protein [K] [CO1,C24]

The adverse effects associated with long-term high protein/high meat intake in humans were (a) disorders of bone and calcium homeostasis, (b) disorders of renal function, (c) increased cancer risk, (d) disorders of liver function, and (e) precipitated progression of coronary artery disease.

8.How would you suggest incorporating fortified foods into a child's diet to prevent nutrient deficiencies? [A] [CO2,CO3,CO4]

common staples like milk, cereals, and cooking oils that have added nutrients like vitamin D, B vitamins, and iron, and serve them to your child. You can also use fortified baby cereals and purees when introducing solids, and select fortified plant-based milks or yogurts

9.How can herbal nutraceuticals affect digestion? [U] [CO4]

Herbal nutraceuticals can affect digestion through a variety of mechanisms, including modulating gut microbiota, stimulating digestive secretions, reducing inflammation, soothing the gut lining, and influencing gastrointestinal motility.

10.Why is water considered as universal solvent? [U]/ [CO2,CO3]

Water is called the universal solvent because it can dissolve a wider variety of substances than any other solvent. This is due to its unique polar molecular structure, where the positive and negative ends of water molecules can attract and pull apart other molecules or ions, allowing them to be dissolved.

11.What are the two common symptoms of iron deficiency anemia?

. [U] [CO2,CO3]

Two common symptoms of iron deficiency anemia are fatigue and shortness of breath. This happens because low iron levels mean the body cannot produce enough hemoglobin to carry oxygen to muscles and tissues, leading to extreme tiredness and a struggle to breathe

12.What is the function of vitamin K? [K] [CO2,CO3]

Vitamin K is essential for the synthesis of proteins needed to stop bleeding, such as prothrombin. It also helps make proteins that play a role in building and maintaining strong bones.

(5 x 2 = 10)

Part C

Short Essay Questions
Answer any **Five** questions
Each question carries **4** marks

13.What role do functional foods play in supporting the immune system. [U] [CO2, CO3,CO4]

Functional foods support the immune system by providing essential nutrients and bioactive compounds that modulate immune responses, protect against pathogens, and improve the body's defense mechanisms. They achieve this through various mechanisms, including antioxidant

properties, the regulation of immune cell activity, and the maintenance of a healthy gut microbiome.

14.Explain the role of dairy products as functional foods. [U] [CO3, CO4]

Health benefits of dairy as functional foods

- Digestive health: Fermented dairy products like yogurt, kefir, and some cheeses contain probiotics that support gut health and can improve digestive function.
- Bone health: Dairy is a rich source of highly absorbable calcium, protein, phosphorus, and vitamin D, all essential for building and maintaining strong bones and potentially preventing osteoporosis.
- Cardiovascular health: Certain compounds in dairy, like bioactive peptides found in fermented milk and whey, have been shown to help lower blood pressure and reduce the absorption of cholesterol.
- Immune support: Dairy products and their components, like whey proteins, can have immunomodulatory effects, helping to regulate immune responses and potentially reducing inflammation

15.Illustrate various adulterants used in food. [U] [CO1,CO2]

Food Item	Common Adulterants	Harmful Effects
Milk	Water, urea, starch, caustic soda, detergents	Stomach disorders, kidney failure, liver and skin disorders
Spices (Chili, Turmeric)	Brick powder, lead chromate, metanil yellow,	Stomach disorders, cancer, liver damage, metal toxicity
Sweets	Metanil yellow, starch, artificial colors	Cancer, stomach disorders, vomiting

16.State the primary functions of fat in the human body. [U] [CO1,CO2]

Function	Description
Energy	Fats are the body's most concentrated energy reserve, stored for use when

Storage	needed, such as during prolonged exercise.
Insulation	A layer of fat beneath the skin helps insulate the body, helping to maintain a stable internal temperature.
Organ Protection	Fat deposits surround and cushion vital organs, protecting them from physical shock and injury.

17.Explain the significance of mineral balance in the body and how deficiencies or excess can affect health. [U] [CO2,CO3]

Significance of mineral balance

- Structural Support: Minerals like calcium are essential for building and maintaining strong bones and teeth.
- Nerve and Muscle Function: Minerals facilitate nerve signal transmission and muscle contractions.
- Fluid and Electrolyte Balance: Minerals such as sodium, potassium, and chloride are electrolytes that help regulate the body's fluid balance and blood pressure.

Health consequences of deficiency

- Osteoporosis: Long-term calcium deficiency can lead to decreased bone density, which can progress to osteoporosis.
- Anemia: Iron deficiency can cause anemia, a condition where the blood can't carry enough oxygen to the body's tissues.
- Impaired Immune Function: Deficiencies in minerals like zinc can lead to a weaker immune system, making you more susceptible to illness

18. Explain the role of lipids in the body, including their function in energy storage, insulation and the absorption of fat-soluble vitamins. [K] [CO4]

Energy storage

- Dense energy source: Lipids provide more energy per gram than carbohydrates, making them an efficient way to store energy.

Insulation and protection

- Thermal insulation: A layer of adipose tissue, a type of lipid, just under the skin acts as an insulating layer to help maintain a stable internal body temperature.
- Organ protection: Lipids cushion and protect vital organs from physical shock and injury.

Absorption of fat-soluble vitamins

- Facilitating absorption: Dietary lipids are essential for the absorption of fat-soluble vitamins A, D, E, and K

19.What are the different types of fats and how do they affect health and energy levels? [K]/ [CO1,CO2]

Types of fats

1.Unsaturated fats: These are healthy fats and are typically liquid at room temperature.

Monounsaturated: Found in olive oil, canola oil, avocados, peanuts, and nuts like almonds and cashews.

Polyunsaturated: Includes omega-3s and omega-6s. Found in sunflower, corn, soybean, and flaxseed oils, as well as fatty fish like salmon and walnuts.

2.Saturated fats: These are typically solid at room temperature and should be consumed in moderation. Sources include fatty cuts of meat, full-fat dairy products, butter, lard, and tropical oils like coconut and palm oil.

3.Trans fats: These are considered unhealthy fats . These are created through hydrogenation, they are found in some margarines, shortening, and commercially baked or fried foods.

(5 x 4 = 20)